

■ Python Practice Problems

Level: **MEDIUM**

30

Problems

Topics covered: **Lists • Tuples • Dictionaries • Functions • Loops • Conditions**

Practice • Learn • Master Python

Table of Contents

- Problem 36.** Two Sum — List | Dict
- Problem 37.** Group Anagrams — Dict | List | Function
- Problem 38.** Longest Consecutive Sequence — List | Dict | Loop
- Problem 39.** Valid Parentheses — String | List (Stack) | Condition
- Problem 40.** Product of Array Except Self — List | Loop
- Problem 41.** Top K Frequent Elements — Dict | List | Function
- Problem 42.** Longest Substring Without Repeating Chars — String | Dict | Loop
- Problem 43.** Subarray Sum Equals K — List | Dict | Loop
- Problem 44.** Sort Dictionary by Value — Dict | Function
- Problem 45.** Matrix Transpose — List | Loop | Function
- Problem 46.** Decode the Message — Dict | String | Function
- Problem 47.** Sliding Window Maximum — List | Loop | Condition
- Problem 48.** Unique Pairs — List | Loop | Tuple
- Problem 49.** Nested Dict Access — Dict | Recursion | Function
- Problem 50.** Frequency Sort — List | Dict | Function
- Problem 51.** Partition List — List | Loop | Condition
- Problem 52.** Dict Difference — Dict | Function
- Problem 53.** Pascal's Triangle Row — List | Loop | Function
- Problem 54.** Spiral Matrix — List | Loop
- Problem 55.** Max Profit (Stock) — List | Loop | Condition
- Problem 56.** Rotate Matrix 90° — List | Loop | Function
- Problem 57.** Tuple Records Filter — Tuple | List | Condition | Function
- Problem 58.** Implement Counter — Dict | Function
- Problem 59.** Continuous Subarray — List | Loop
- Problem 60.** Dictionary Comprehension Filter — Dict | Condition | Function
- Problem 61.** Balanced Brackets Check — String | List | Dict
- Problem 62.** Nested List Sum — List | Recursion | Function
- Problem 63.** Moving Average — List | Loop | Function
- Problem 64.** Convert List to Dict Tree — List | Dict | Loop
- Problem 65.** Chunk a List — List | Loop | Function

Problem 36

Two Sum

Tags: List | Dict

Given a list of integers **nums** and an integer **target**, return indices of the two numbers that add up to **target**. Each input has exactly one solution; you may not use the same element twice.

Example 1:

Input: `nums=[2,7,11,15], target=9`

Output: `[0,1]`

Explanation: `nums[0]+nums[1]=9`

Example 2:

Input: `nums=[3,2,4], target=6`

Output: `[1,2]`

Example 3:

Input: `nums=[3,3], target=6`

Output: `[0,1]`

Problem 37

Group Anagrams

Tags: Dict | List | Function

Given a list of strings **strs**, group the anagrams together. You can return the answer in any order.

Example 1:

Input: `strs=["eat","tea","tan","ate","nat","bat"]`

Output: `[["bat"],["nat","tan"],["ate","eat","tea"]]`

Example 2:

Input: `strs=[]`

Output: `[[""]]`

Example 3:

Input: `strs=["a"]`

Output: `[["a"]]`

Problem 38

Longest Consecutive Sequence

Tags: List | Dict | Loop

Given a list of integers **nums**, return the length of the longest consecutive elements sequence. Your solution should run in $O(n)$ time.

Example 1:

Input: `nums=[100,4,200,1,3,2]`

Output: 4

Explanation: `[1,2,3,4]`

Example 2:

Input: `nums=[0,3,7,2,5,8,4,6,0,1]`

Output: 9

Example 3:

Input: `nums=[]`

Output: 0

Problem 39

Valid Parentheses

Tags: String | List (Stack) | Condition

Given a string **s** containing only '(', ')', '{', '}', '[', ']', determine if the input string is valid. Open brackets must be closed in the correct order.

Example 1:

Input: `s="()"`

Output: True

Example 2:

Input: `s="()[]{}"`

Output: True

Example 3:

Input: `s="]"`

Output: False

Problem 40

Product of Array Except Self

Tags: List | Loop

Given an integer list **nums**, return a list **output** where **output[i]** is the product of all elements of **nums** except **nums[i]**. Do not use division.

Example 1:

Input: `nums=[1, 2, 3, 4]`

Output: `[24, 12, 8, 6]`

Example 2:

Input: `nums=[ -1, 1, 0, -3, 3]`

Output: `[0, 0, 9, 0, 0]`

Example 3:

Input: `nums=[2, 3]`

Output: `[3, 2]`

Problem 41

Top K Frequent Elements

Tags: Dict | List | Function

Given an integer list **nums** and an integer **k**, return the **k** most frequent elements in any order.

Example 1:

Input: `nums=[1, 1, 1, 2, 2, 3], k=2`

Output: `[1, 2]`

Example 2:

Input: `nums=[1], k=1`

Output: `[1]`

Example 3:

Input: `nums=[1, 2, 2, 3, 3, 3], k=1`

Output: `[3]`

Problem 42

Longest Substring Without Repeating Chars

Tags: String | Dict | Loop

Given a string **s**, find the length of the longest substring without repeating characters.

Example 1:

Input: s="abcabcbb"

Output: 3

Explanation: 'abc' has length 3

Example 2:

Input: s="bbbbbb"

Output: 1

Example 3:

Input: s="pwwkew"

Output: 3

Explanation: 'wke' has length 3

Problem 43

Subarray Sum Equals K

Tags: List | Dict | Loop

Given an integer list **nums** and integer **k**, return the total number of subarrays whose sum equals **k**.

Example 1:

Input: nums=[1,1,1], k=2

Output: 2

Example 2:

Input: nums=[1,2,3], k=3

Output: 2

Example 3:

Input: nums=[-1,-1,1], k=0

Output: 1

Problem 44

Sort Dictionary by Value

Tags: Dict | Function

Given a dictionary **d**, return a new dictionary sorted by its values in descending order.

Example 1:

Input: `d={'apple':3, 'banana':1, 'cherry':2}`

Output: `{'apple':3, 'cherry':2, 'banana':1}`

Example 2:

Input: `d={'a':10, 'b':10, 'c':5}`

Output: `{'a':10, 'b':10, 'c':5}`

Example 3:

Input: `d={'x':1}`

Output: `{'x':1}`

Problem 45

Matrix Transpose

Tags: List | Loop | Function

Given a 2D list (matrix) **matrix**, return its transpose. The transpose of a matrix is the matrix flipped over its diagonal.

Example 1:

Input: `matrix=[[1,2,3],[4,5,6],[7,8,9]]`

Output: `[[1,4,7],[2,5,8],[3,6,9]]`

Example 2:

Input: `matrix=[[1,2],[3,4],[5,6]]`

Output: `[[1,3,5],[2,4,6]]`

Example 3:

Input: `matrix=[[1]]`

Output: `[[1]]`

Problem 46

Decode the Message

Tags: Dict | String | Function

Given a cipher mapping dictionary **cipher** and an encoded string **msg**, decode each character using the cipher. Characters not in cipher remain unchanged.

Example 1:

```
Input: cipher={'a':'z', 'b':'y'}, msg='abc'
```

```
Output: 'zyc'
```

Example 2:

```
Input: cipher={'1':'a'}, msg='12'
```

```
Output: 'a2'
```

Example 3:

```
Input: cipher={}, msg='hello'
```

```
Output: 'hello'
```

Problem 47

Sliding Window Maximum

Tags: List | Loop | Condition

Given a list **nums** and integer **k**, return a list of the maximum values in each sliding window of size **k**.

Example 1:

```
Input: nums=[1,3,-1,-3,5,3,6,7], k=3
```

```
Output: [3,3,5,5,6,7]
```

Example 2:

```
Input: nums=[1], k=1
```

```
Output: [1]
```

Example 3:

```
Input: nums=[1,3,1], k=3
```

```
Output: [3]
```


Problem 48

Unique Pairs

Tags: List | Loop | Tuple

Given a list of integers **nums** and a target **target**, return all unique pairs of numbers that sum to **target** as a list of tuples.

Example 1:

```
Input: nums=[1,2,3,4,5], target=6
```

```
Output: [(1,5),(2,4)]
```

Example 2:

```
Input: nums=[1,1,2,3], target=4
```

```
Output: [(1,3)]
```

Example 3:

```
Input: nums=[0,0,0], target=0
```

```
Output: [(0,0)]
```

Problem 49

Nested Dict Access

Tags: Dict | Recursion | Function

Given a nested dictionary **d** and a list of keys **path**, return the value found by following the path. Return **None** if the path doesn't exist.

Example 1:

```
Input: d={'a':{'b':{'c':42}}}, path=['a','b','c']
```

```
Output: 42
```

Example 2:

```
Input: d={'a':1}, path=['a','b']
```

```
Output: None
```

Example 3:

```
Input: d={'x':10}, path=['x']
```

```
Output: 10
```

Problem 50

Frequency Sort

Tags: List | Dict | Function

Given a list **nums**, sort it so that elements with higher frequency come first. For ties in frequency, smaller numbers come first.

Example 1:

Input: `nums=[1,1,2,2,2,3]`

Output: `[2,2,2,1,1,3]`

Example 2:

Input: `nums=[2,3,1,3,2]`

Output: `[2,2,3,3,1]`

Example 3:

Input: `nums=[-1,1,-6,4,5,-6,1,4,1]`

Output: `[1,1,1,-6,-6,4,4,-1,5]`

Problem 51

Partition List

Tags: List | Loop | Condition

Given a list **nums** and integer **pivot**, rearrange the list so all elements less than **pivot** come first, then elements equal to **pivot**, then greater. Preserve relative order within each group.

Example 1:

Input: `nums=[3,1,4,1,5,9,2,6], pivot=4`

Output: `[3,1,1,2,4,5,9,6]`

Example 2:

Input: `nums=[1,2,3], pivot=2`

Output: `[1,2,3]`

Example 3:

Input: `nums=[5,5,5], pivot=5`

Output: `[5,5,5]`

Problem 52

Dict Difference

Tags: Dict | Function

Given two dictionaries **d1** and **d2**, return a dictionary of keys that appear in **d1** but not in **d2**, along with their values from **d1**.

Example 1:

Input: d1={'a':1, 'b':2, 'c':3}, d2={'b':9, 'c':8}

Output: {'a':1}

Example 2:

Input: d1={'x':1}, d2={'x':1}

Output: {}

Example 3:

Input: d1={'a':1}, d2={}

Output: {'a':1}

Problem 53

Pascal's Triangle Row

Tags: List | Loop | Function

Given an integer **rowIndex** (0-indexed), return the **rowIndex**-th row of Pascal's triangle.

Example 1:

Input: rowIndex=3

Output: [1, 3, 3, 1]

Example 2:

Input: rowIndex=0

Output: [1]

Example 3:

Input: rowIndex=4

Output: [1, 4, 6, 4, 1]

Problem 54

Spiral Matrix

Tags: List | Loop

Given an $m \times n$ matrix, return all elements in spiral order (clockwise from top-left).

Example 1:

Input: `matrix=[[1,2,3],[4,5,6],[7,8,9]]`

Output: `[1,2,3,6,9,8,7,4,5]`

Example 2:

Input: `matrix=[[1,2],[3,4]]`

Output: `[1,2,4,3]`

Example 3:

Input: `matrix=[[1]]`

Output: `[1]`

Problem 55

Max Profit (Stock)

Tags: List | Loop | Condition

Given a list **prices** where **prices[i]** is the stock price on day **i**, return the maximum profit you can achieve from a single buy-sell. Return 0 if no profit is possible.

Example 1:

Input: `prices=[7,1,5,3,6,4]`

Output: 5

Explanation: Buy at 1, sell at 6

Example 2:

Input: `prices=[7,6,4,3,1]`

Output: 0

Explanation: Prices always fall

Example 3:

Input: `prices=[1,2]`

Output: 1

Problem 56

Rotate Matrix 90°

Tags: List | Loop | Function

Given an $n \times n$ 2D matrix, rotate it 90 degrees clockwise in-place.

Example 1:

Input: `matrix=[[1,2,3],[4,5,6],[7,8,9]]`

Output: `[[7,4,1],[8,5,2],[9,6,3]]`

Example 2:

Input: `matrix=[[5,1,9,11],[2,4,8,10],[13,3,6,7],[15,14,12,16]]`

Output: `[[15,13,2,5],[14,3,4,1],[12,6,8,9],[16,7,10,11]]`

Example 3:

Input: `matrix=[[1]]`

Output: `[[1]]`

Problem 57

Tuple Records Filter

Tags: Tuple | List | Condition | Function

Given a list of tuples **records** where each tuple is (name, score, passed), return a list of names of students who passed (passed=True) and scored above **threshold**.

Example 1:

Input: `records=[('Ali',85,True),('Bob',60,False),('Eva',90,True)], threshold=80`

Output: `['Ali','Eva']`

Example 2:

Input: `records=[('X',50,True)], threshold=60`

Output: `[]`

Example 3:

Input: `records=[], threshold=70`

Output: `[]`

Problem 58

Implement Counter

Tags: Dict | Function

Implement a function **counter(items)** that returns a dictionary with the count of each element in the iterable **items** (similar to `collections.Counter`), without using `Counter`.

Example 1:

```
Input: items=[1,2,2,3,3,3]
```

```
Output: {1:1, 2:2, 3:3}
```

Example 2:

```
Input: items='mississippi'
```

```
Output: {'m':1, 'i':4, 's':4, 'p':2}
```

Example 3:

```
Input: items=[]
```

```
Output: {}
```

Problem 59

Continuous Subarray

Tags: List | Loop

Given an integer list **nums**, find the contiguous subarray (containing at least one number) which has the largest sum and return its sum (Kadane's Algorithm).

Example 1:

```
Input: nums=[-2,1,-3,4,-1,2,1,-5,4]
```

```
Output: 6
```

Explanation: [4,-1,2,1]

Example 2:

```
Input: nums=[1]
```

```
Output: 1
```

Example 3:

```
Input: nums=[5,4,-1,7,8]
```

```
Output: 23
```

Problem 60

Dictionary Comprehension Filter

Tags: Dict | Condition | Function

Given a dictionary **d** and integer **threshold**, return a new dictionary containing only key-value pairs where the value is greater than **threshold**.

Example 1:

Input: `d={'a':10, 'b':3, 'c':7}, threshold=5`

Output: `{'a':10, 'c':7}`

Example 2:

Input: `d={'x':1, 'y':2}, threshold=5`

Output: `{}`

Example 3:

Input: `d={'m':100}, threshold=0`

Output: `{'m':100}`

Problem 61

Balanced Brackets Check

Tags: String | List | Dict

Given a string **s** with mixed characters, check if all bracket characters (`()[]{}`) are balanced. Non-bracket characters should be ignored.

Example 1:

Input: `s="a(b[c]d)e"`

Output: `True`

Example 2:

Input: `s="a(b[c)d]e"`

Output: `False`

Example 3:

Input: `s="no brackets"`

Output: `True`

Problem 62

Nested List Sum

Tags: List | Recursion | Function

Given a nested list **nested** (lists within lists, arbitrary depth), return the sum of all integers.

Example 1:

Input: `nested=[1, [2, [3, 4]], 5]`

Output: 15

Example 2:

Input: `nested=[[1, 2], [3, [4, [5]]]]`

Output: 15

Example 3:

Input: `nested=[1, 2, 3]`

Output: 6

Problem 63

Moving Average

Tags: List | Loop | Function

Given a list of numbers **nums** and window size **k**, return a list of moving averages for each window of size **k**.

Example 1:

Input: `nums=[1, 2, 3, 4, 5], k=3`

Output: `[2.0, 3.0, 4.0]`

Example 2:

Input: `nums=[10, 20, 30], k=2`

Output: `[15.0, 25.0]`

Example 3:

Input: `nums=[1, 2, 3, 4], k=4`

Output: `[2.5]`

Problem 64

Convert List to Dict Tree

Tags: List | Dict | Loop

Given a list of (parent, child) pairs **edges**, build a dictionary representing the tree where each key maps to a list of its children.

Example 1:

Input: edges=[('A', 'B'), ('A', 'C'), ('B', 'D')]

Output: {'A': ['B', 'C'], 'B': ['D'], 'C': [], 'D': []}

Example 2:

Input: edges=[]

Output: {}

Example 3:

Input: edges=[('root', 'leaf')]

Output: {'root': ['leaf'], 'leaf': []}

Problem 65

Chunk a List

Tags: List | Loop | Function

Given a list **nums** and chunk size **n**, split the list into sublists of size **n**. The last sublist may be smaller.

Example 1:

Input: nums=[1, 2, 3, 4, 5, 6, 7], n=3

Output: [[1, 2, 3], [4, 5, 6], [7]]

Example 2:

Input: nums=[1, 2, 3, 4], n=2

Output: [[1, 2], [3, 4]]

Example 3:

Input: nums=[], n=5

Output: []